27/9/26

Practical - 9

Subnetting in Cisco Packet Tracer Simulator

Aim:

Implementation of Subnetting in Cisco Packet Tracer Simulator.

Classless IP subnetting is a technique that allows for more efficient use of IP addresses by allowing for subnet masks that are not just the default masks for each IP class. This means that we can divide our IP address ~~spce~~ space into smaller subnets, which can be useful when we have a limited number of IP addresses but need to create multiple networks.

Creating a Network Topology:

The first step ~~is~~ in implementing classless IP subnetting is to create a network topology in Packet Tracer. To create a network topology in Packet Tracer, select the "New" button in the top left corner, then select "Network" and "Generic". This will create a blank network topology that we can use to add devices.

Adding The Devices:

Once we have created our network topology, we can add devices to it. Here, we will be adding routers, switches, and PCs. To add a device, select the device from the bottom left corner and drag it onto the network topology. Then, connect the devices by dragging a cable from one device's port to another device's port.

Subnetting:

To subnet the network address of 192.168.1.0/24 to provide enough space for atleast 5 addresses for end devices,

the switch, and the router, we can use a /27 subnet mask. This will give us 8 subnets with 30 host addresses each.

The IP addressing for the network shown in the topology can be as follows:

Router R1:
GigabitEthernet 0/0: 192.168.1.1
GigabitEthernet 0/1: 192.168.2.1
Switch S1:
FastEthernet 0/1: 192.168.1.0/27
PC1: 192.168.1.11
PC2: 192.168.1.12
PC3: 192.168.1.13
PC4: 192.168.1.14
PC5: 192.168.1.15
FastEthernet 0/2: 192.168.2.6/27
PC1: 192.168.2.11
PC2: 192.168.2.12
PC3: 192.168.2.13
PC4: 192.168.2.14
PC5: 192.168.2.15
Router R2:
FastEthernet 0/0: 192.168.3.1
FastEthernet 0/1: 192.168.4.1
Switch S2
FastEthernet 0/1: 192.168.3.0/27
PC1: 192.168.3.11
PC2: 192.168.3.12
PC3: 192.168.3.13
PC4: 192.168.3.14
PC5: 192.168.3.15

FastEthernet0/2: 192.168.4.0/27
PC1: 192.168.4.11
PC2: 192.168.4.12
PC3: 192.168.4.13
PC4: 192.168.4.14
PC5: 192.168.4.15

Configuring the Devices:
Now that we have added our devices and connected them, we can start configuring them. We will start by configuring the router. Right-click on the router and select CLI. This will open the command-line interface for the router. In the CLI, enter the following commands:

```
#enable
#configure terminal
#interface FastEthernet0/0
#ip address {IP address} {subnet mask}
#no shutdown
#exit
interface FastEthernet0/1
ip address {IP address} {subnet mask}
no shutdown
exit.
```

Replace "{IP address}" and "{subnet mask}" with your desired IP address and subnet mask. The first interface FastEthernet0/0 will be connected to the switch, while the second interface, FastEthernet0/1 will be connected to the switch, while the second interface, FastEthernet0/0 will be connected to PC5. These commands configure the router's interfaces with IP addresses and subnet masks.
Next we will configure the switch. Right-click on the switch and select CLI. In the CLI enter the following commands.

enable
configure terminal
interface FastEthernet 0/1
switchport mode access
exit

interface FastEthernet 0/2
switchport mode access
exit

These commands configure the switch to operate in access mode on its two ports, which are connected to the two PCs.
Finally, we will configure the PCs. Right-Click on each PC and select "Config". In the configuration window, enter the IP address, subnet mask, default gateway and DNS server information. The IP address and subnet mask should be within the same subnet as the router's FastEthernet 0/1 interface.
To configure the GigabitEthernet interface on the router, you can follow these steps.
1. Right-click on the router and select "CLI"
2. Enter the following commands

enable
configure terminal
interface GigabitEthernet 0/0
ip address {IP address} {subnet mask}
no shutdown
exit.

Replace "{IP address}" and "{subnet mask}" with your ~~desired~~ desired IP address and subnet mask. These commands configure the Gigabit Ethernet interface with an IP address and

subnet mask and enable the interface.

Testing the Network:

Now that our network topology is configured we can test the network. Open a command prompt on each PC and try to ping the other PC. If the ping is successful, then the network is functioning properly. We can also use the ping command to test connectivity between the router and the PCs.

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.1.34    YES manual up                    up
GigabitEthernet0/0/1   192.168.1.65    YES manual up                    up
Router#
```

```
C:\>ping 192.168.1.71

Pinging 192.168.1.71 with 32 bytes of data:

Reply from 192.168.1.71: bytes=32 time<1ms TTL=126
Reply from 192.168.1.71: bytes=32 time<1ms TTL=126
Reply from 192.168.1.71: bytes=32 time<1ms TTL=126
Reply from 192.168.1.71: bytes=32 time<1ms TTL=126

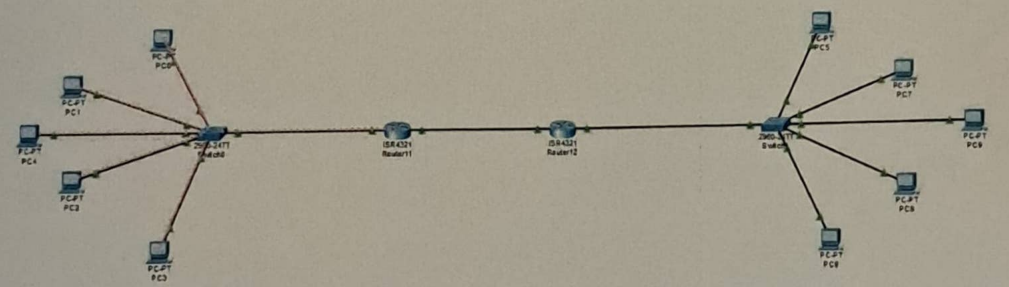
Ping statistics for 192.168.1.71:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```





Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete	
	Successful	PC4	PC5	ICMP		0.000	N	0	(edit)		(delete)
	Successful	PC0	PC3	ICMP		0.000	N	1	(edit)		(delete)

Result:
Thus subnetting is implemented in Cisco Packet Tracer Simulator.